Linguistic Interpretability

Natural language processing

Student: Ali Dankoob

40407024

Linguistic Interpretability of Transformer-based PLMs

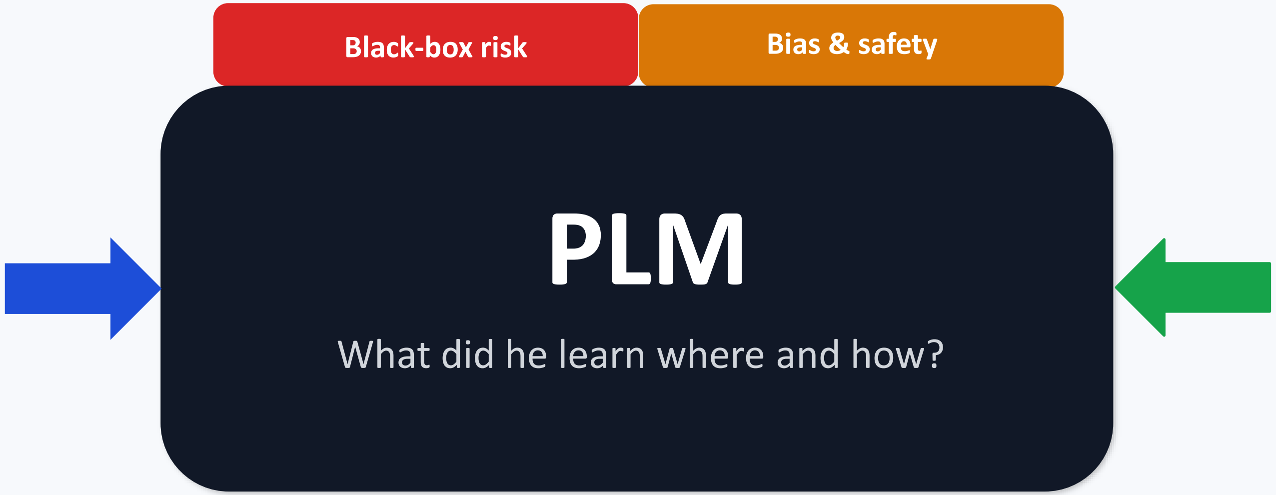
Systematic Review of 160 Articles + Roadmap (Prototype)

- Problem and Motivation: Why is Linguistic Interpretability Important?
- Systematic Review Method and Scope of the Article (PRISMA-like)
- Overview of Transformer and the Concept of layerwise interpretability
- Key results at 4 levels of Syntax/Semantics/Morphology/Discourse:
- Methods: Probing, Analysis, Attention, Geometry Analysis, Embedding, etc.
- Limitations + Prototype Idea Practical for Lesson Project

1)Problem and motivation

Why Interpretability at all?

- Transformer models perform great, but there's a "black box" inside.
- Without an understanding of the internal mechanism: it becomes difficult to detect errors, biases, and fragile behaviors.
- For practical NLP: Knowing which layer/head/neuron has what knowledge debugging and targeted improvement.
- For low-resource languages: If we understand where linguistic knowledge is encoded, it becomes easier to transfer/inject knowledge.



Why is "linguistics" within the model important?

- If the model really learns linguistic structures, we expect better generalization and fewer errors.
- If it only learns statistical correlation, it may remain dependent on frequency.
- Objective: To summarize the evidence from 160 research papers on Syntax/Morphology/Semantics/Discourse in PLMs.

2) Definitions and Scope

Three Keywords

Explainability

Explain why this output is usually close to the output/behavior? It is often not model-agnostic and focuses on the importance of inputs.

Interpretability

Trying to understand the internal mechanisms (layers, attention heads, neurons, embeddings) and how they reach the output.

Linguistic interpretability

Focus on whether PLMs represent/code linguistic phenomena (syntax/morphology/semantics/discourse) similar to humans.

What exactly does this survey cover?

- 160 papers (2018 to Oct 2024), focusing on Transformer-based PLMs and internal representations.
- Baseline models (pre-trained) without fine-tune for specific tasks; only black-box studies have been omitted
- The four linguistic levels are Syntax, Semantics, Morphology, and Discourse
- Methods: Probing Attention/Neurons Analysis, Embedding Geometry Analysis, Ablation/Pruning, etc.

Selection of Pipeline Articles (Key Numbers)

Google Scholar search
≈ 5,580 results

After filters & sampling
876 candidate papers

Title/abstract screening
277 papers

Full analysis + exclusions
160 included studies

Delete/Important Details in Selection

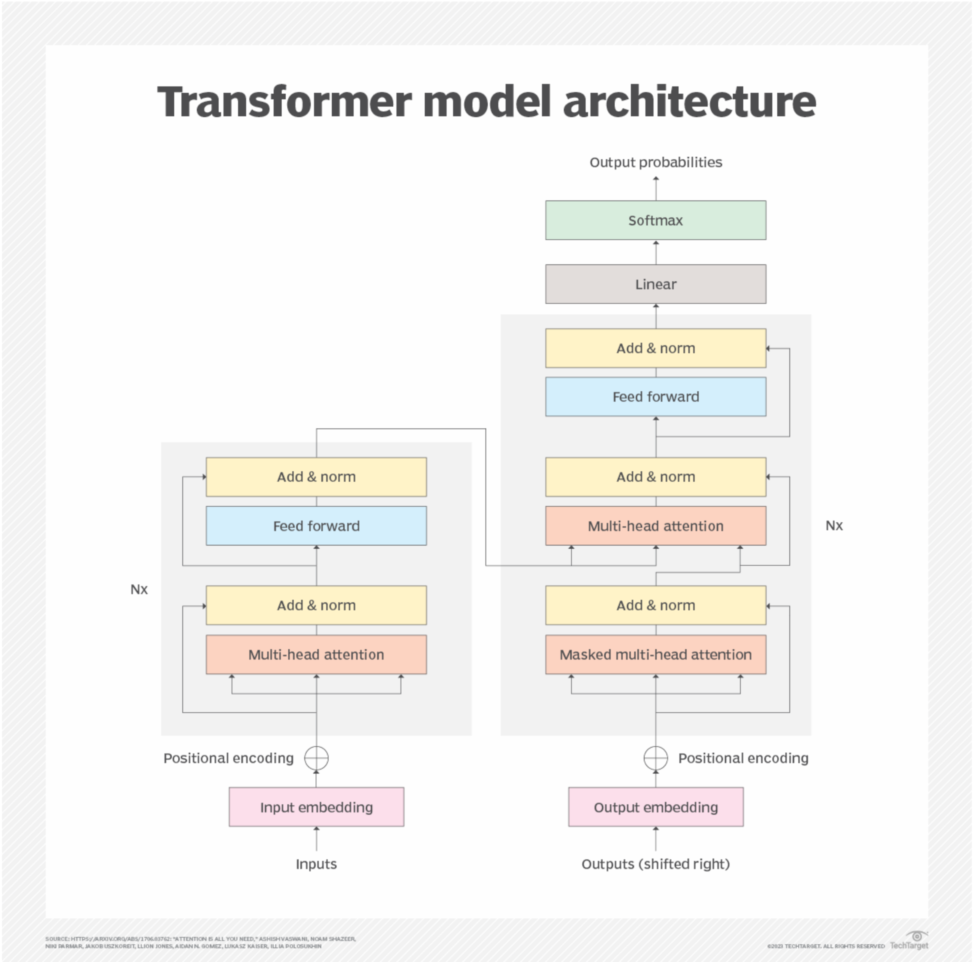
- Compound keywords : probe/interpret/explain+ transformer/BERT +syntax/semantic/morphology...
- Filter: Period 2018–Oct 2024, focusing on English articles and mainly peer-reviewed articles
- Eliminate black-box analyses only for the final output; focus on layerwise/internal.
- Two stages of human annotation: (1) title/abstract, (2) accurate reading and dimensional labeling.

Labeling Dimensions: Linguistic Level, Phenomena, Languages, Models, Methodology, Attitude toward PLM's Linguistic Ability, and (interpretability)

- Survey → "What can be put into practice?“
- The prototype must demonstrate an intrinsic interpretability technique (e.g., probing).
- Output: Chart + Analysis + Reproducible Code.

Key components studied in the studies

- Tokenization (BPE/WordPiece) → can make morphology difficult to study.
- Embedding layer + L Transformer layers → contextual embeddings in each layer.
- Self-attention heads → patterns of dependence, coreference, constituency...
- Feed-forward (FFN) + residual/LayerNorm → "store/convert" knowledge in layers.
- Layerwise view: lower (1–3), middle (4–7), upper (8–12) for BERT-base.
- Encoder vs Decoder vs Encoder-Decoder → Difference in Attention (Masking) and Application.



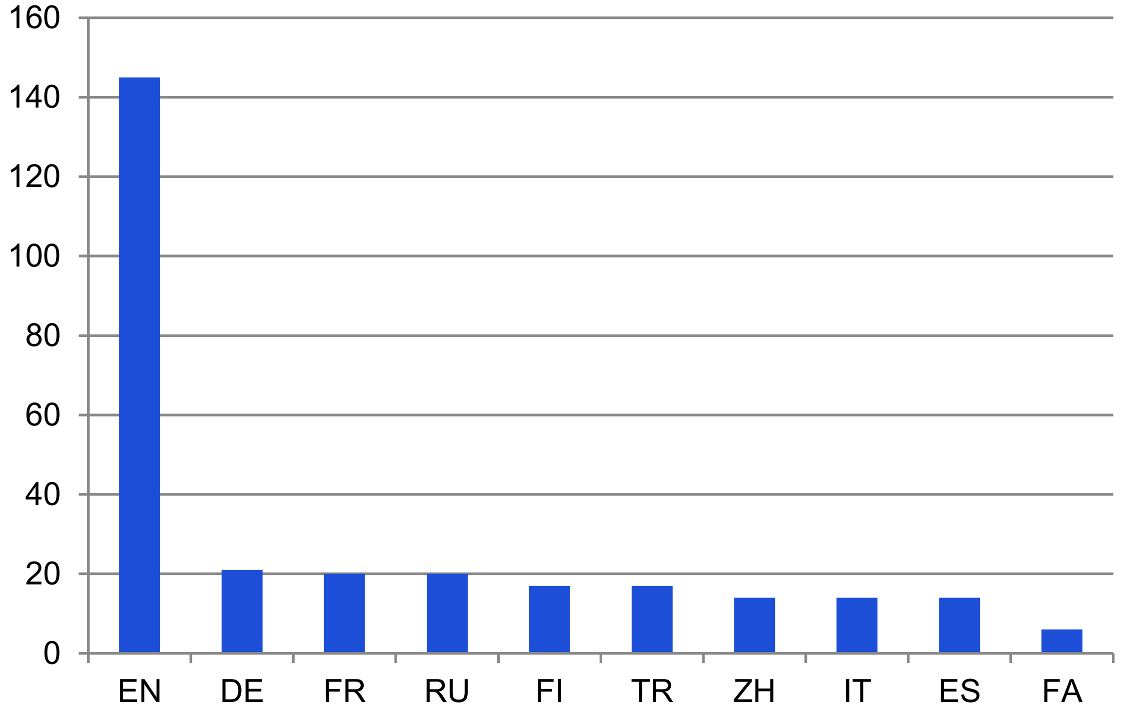
Transformer model architecture (encoder/decoder)

5) Survey Data: Languages and Architectures Surveyed

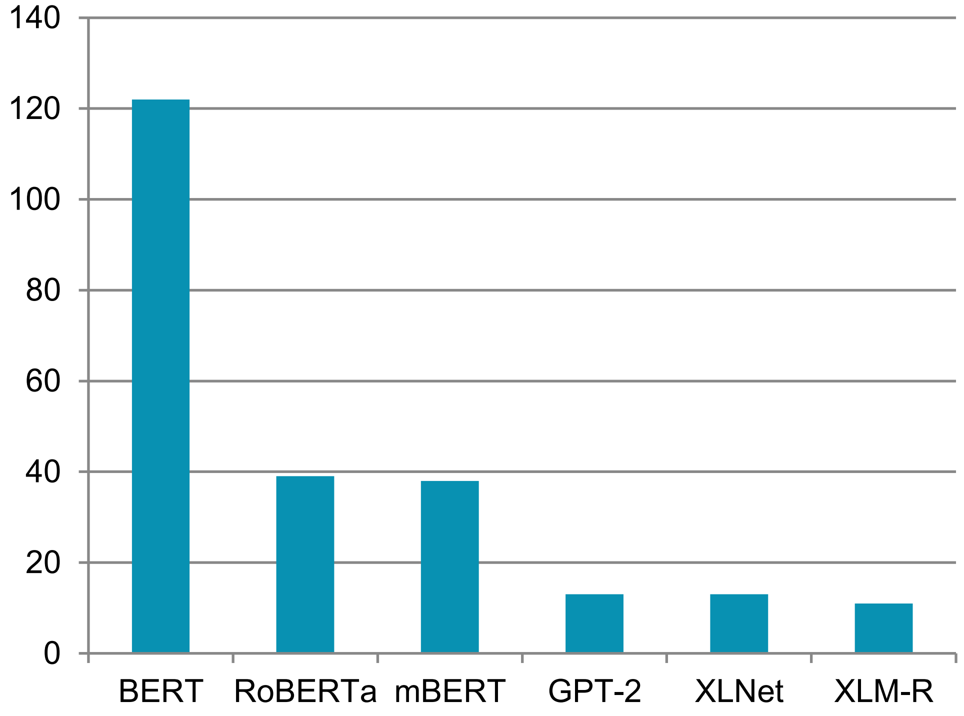
Research focus: Still English-centric

145/160 studies include English, but other languages have been gradually added (e.g., German, French, Russian, Persian...).

Top languages in analyzed PLMs (count of studies)



Most studied model families (count of studies)



Insight: The heavy focus on BERT/RoBERTa is due to its "parameter availability" and lower cost of internal analysis.

4 Family Method (Taxonomy Article)

Feature attribution

The importance of each token/span relative to the output (gradient, perturbation, LIME...).

Example-based

Designed/Adversarial/Counterfactual sentences and view the output change/display.

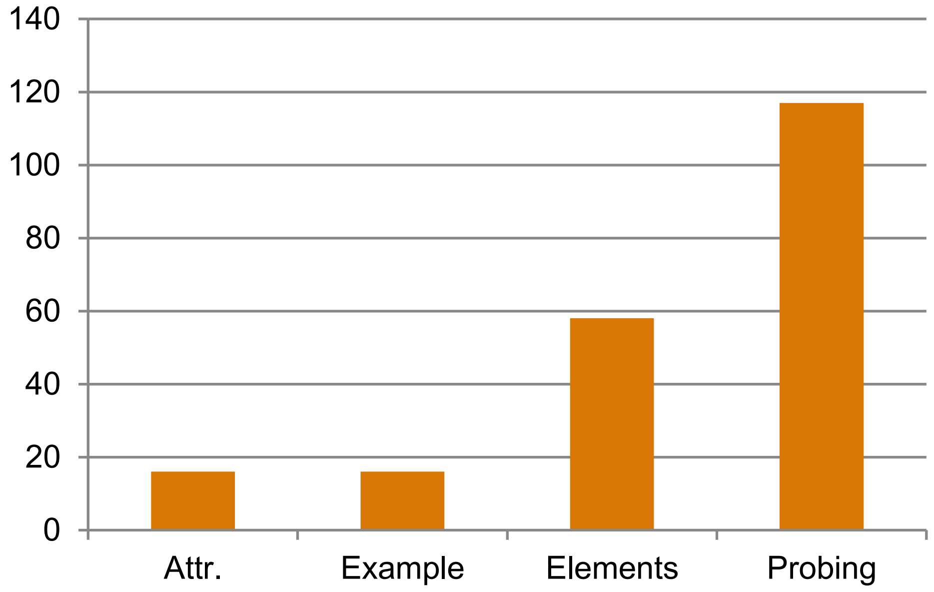
Architecture elements

Analysisattention heads/neurons/FFN/embeddings + pruning/ablation/visualization.

Probing

Style classifier tutorial on intermediate representations to predict language labels (POS, dep, SRL...).

How often each method appears (160 papers)



Quick Withdrawal:

- Probing is the dominant method (117 studies).
- A combination of several methods usually yields richer results.
- Probing risk: "Memorizing the task" rather than recognizing the knowledge of the model → requires control/causal probes.

7) Findings — Syntax

Is syntax in PLMs? Probably yes — but with a caveat.

1) Positive evidence

- Discover dependency-like structure of embeddings (multiple studies).
- Many report that syntax is more powerful than semantics code.
- In mBERT there is evidence of syntactic transfer between languages.

2) Negative/ambiguous evidence

- Dependence on word order in some settings.
- Weakness in Specific Phenomena: Agreement in Low-Frequency Forms, Reflexive Anaphora...
- The syntax/semantics separation is not always clear (inconsistent results).

3) Layerwise & elements

- BERT-base: Often important "middle layers" are reported (but not complete consensus).
- Some tasks: the simultaneous role of lower+middle or distribution across the entire model.
- Attention heads are sometimes specialized in dependency relationships.

4) Embedding geometry

- Separate linear spaces are reported below for syntax and semantics.
- Low-dimensional subspaces are reported for some syntactic variables (e.g., POS/agreement).
- However, overlap between linguistic concepts has also been observed.

8) Findings — (Lexico)Semantics

Overall result: Semantics are largely present, but heterogeneous and sometimes weaker than syntax

Reported capabilities

- Polysemy and sense-level differences (monolingual is better than mBERT in some studies).
- Knowledge of scale/number/date and some semantic roles.
- Recognize the meaning of idioms (literal vs figurative) in some settings.

Challenges and Cons

- Negation: Mixed results; unstable in some models/sizes.
- Context-dependent reasoning/factual abstraction has been reported in some works.
- Domain semantics (e.g., scientific) are sometimes weak and cause chain errors (e.g., coref).

Layers and geometry

- There is no definitive consensus on layerwise location (middle vs upper vs lower depending on the model).
- The lower layers are often lexical.
- The existence of separate subspaces is reported for semantics and syntax.

9) Findings — Morphology

Morphology is less studied, and tokenization is a serious obstacle

Key observations (from the Survey)

- Most studies examine morphology along with syntax (not independent).
- Tokenization (WordPiece/BPE) makes morphemes not always equivalent to tokens.
- Layerwise: Sometimes similar to syntax (middle/upper), but depending on the language and model.
- Some phenomena (number/tense) are reported in linear subspaces.
- Evidence of common 'morphosyntax' neurons in mBERT has been reported between languages.

Implications for Prototype

- If Persian/Turkish/Russian is selected: Be sure to report the sensitivity of the tokenizer.
- For class work: Agreement (number/tense) is very implementable with probing.
- It is possible to specify which layers are better for number/tense.

Sample Suggested Experiment (Quick)

- Dataset: UD POS + Morph features (small subset).
- Extract: embeddings of multiple layers (e.g., 1,4,8,12).
- Probe: Logistic regression/small MLP for features like Number/Tense.
- Output: accuracy per layer + chart + analysis.

10) Discourse and Minor Issues

There is a lot of research that has been done on the subject (6 articles), but it has some interesting findings.

Discourse findings

- Coherence and discourse relationships are seen to some extent in embeddings.
- The sensitivity to disfluency decreases with the depth of the layer (the representations become more similar).
- RST: Some jobs report the intermediate layers more strongly.

Knowledge-domain neurons

- Report the presence of "scoped" neurons (social media / science / noun-heavy ...).
- Discovery by clustering the activation of neurons on sentences.
- For the project: A simple simulation can be done with activation statistics.

Multilingual shared space

- Hypothesis: Common parameters → alignment of structures between languages.
- Mixed evidence: Some phenomena are shared, some are not (e.g., agreement in masked LMs).
- Word order plays a key role in several studies.

review this slide as "Lesser-said but interesting".

Why aren't interpretability results always "proven"?

Correlation ≠ Causation

- Linguistic information may be "extractable" in embedding, but the model may not use it in inference.
- Probes may learn the task rather than recognize the knowledge of the model.
- Solutions: control tasks, causal probes (amnesic/information-theoretic/dropout...).
- Limited metric (accuracy only) can be misleading.

Why have new LLMs become harder to analyze?

- Model size ↑ → Cost of extraction/internal analysis ↑Some models are closed (parameters/heads are not available).
- Tendency to prompting in studies ↑, but transparency/internal analysis ↓
- Conclusion: Research is still very much focused on the BERT-family.

12) Prototype – What Was Implemented

Prototype Overview

- ✓ Model: bert-base-multilingual-cased (Transformer encoder)
- ✓ Data: Universal Dependencies (UD) – POS annotations
- ✓ Method: Layer-wise probing of contextual embeddings
- ✓ Analysis: Extraction of hidden states from multiple Transformer layers
- ✓ Goal: Demonstrate how syntactic information is distributed across layers

- > This prototype is inspired by the dominant interpretability methods identified in the survey.

Prototype Overview

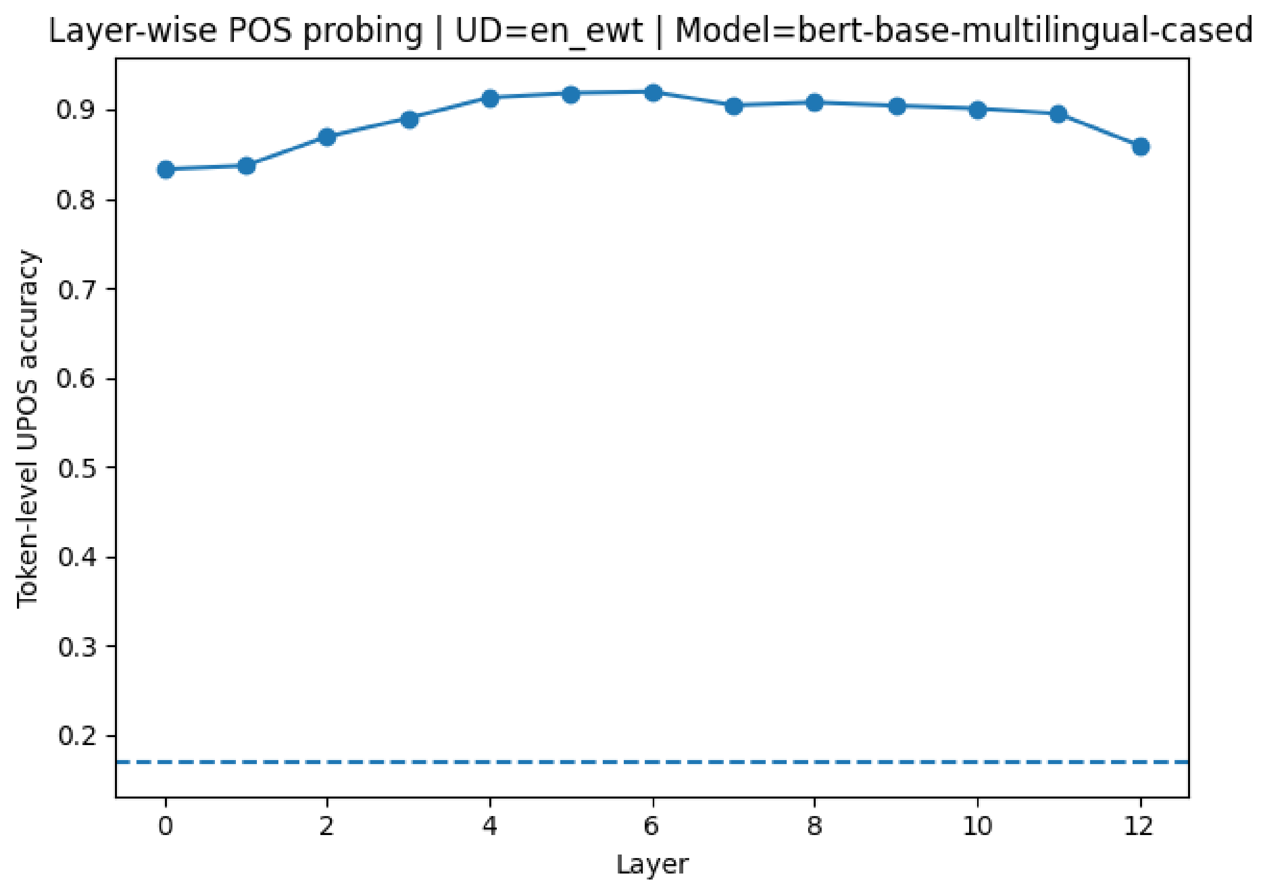
✓ Key Observations

Syntactic information is not uniformly distributed across layers

Middle layers tend to show stronger POS predictability

Lower layers capture more lexical information

Upper layers show more abstract representations



layer-wise accuracy plot

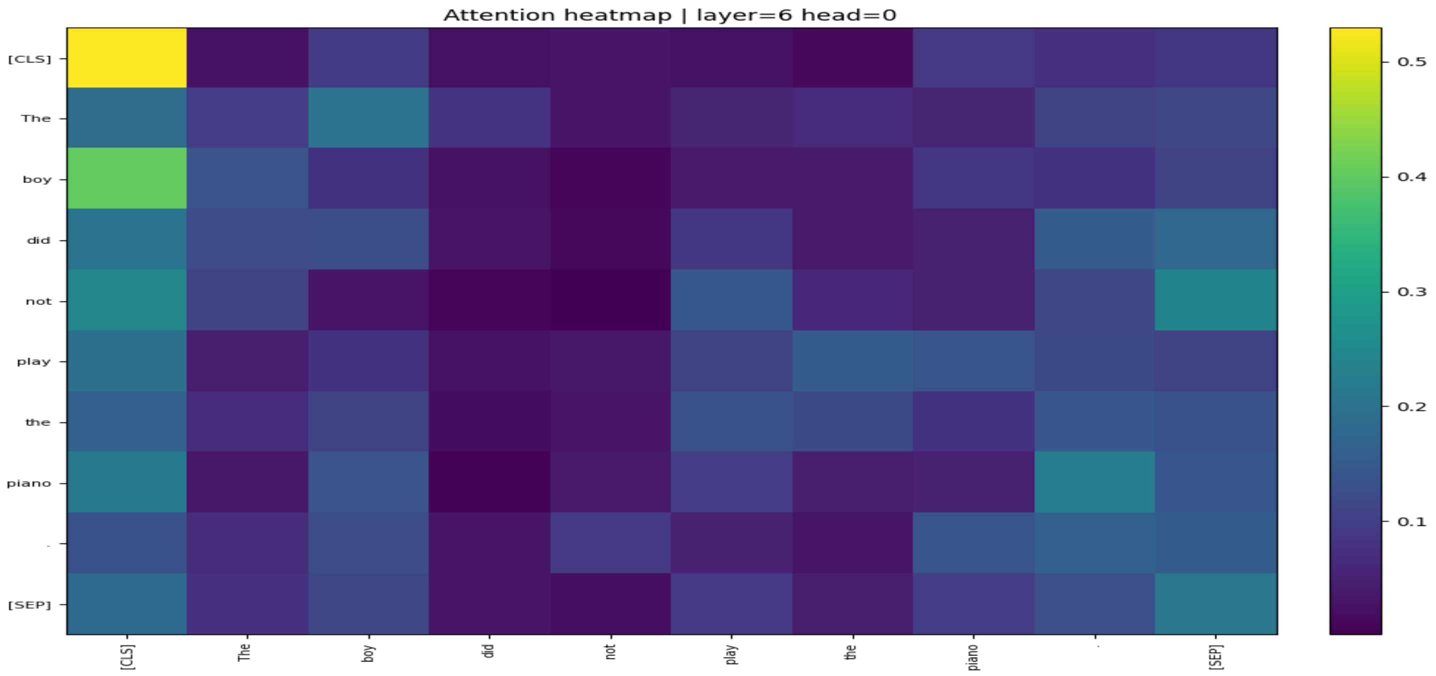
14) Attention Visualization (Example)

Attention-Based Analysis

Data (UD subset) → Tokenizer (BPE/WordPiece) → PLM (bert-base) → Extract layer embeddings → Probe (LogReg/MLP)

attention heatmap image

- Selected attention heads show structured patterns
  Some heads align with syntactic relations (e.g., local dependencies)
  Attention analysis complements probing but does not replace it



15) Limitations & Disclaimer

Why This Prototype Is Valid for the Course (16

- ✓ Important Caveats
- ✓ This is a survey-based prototype, not a replication study
- ✓ Probing measures extractability, not causal usage
- ✓ Results are illustrative and educational
- ✓ Small-scale experiment with limited data and models

Justification

- The reference paper is a survey, not an experimental benchmark ✓
- The prototype directly implements core methods identified in the survey ✓
- Demonstrates practical feasibility of interpretability techniques ✓
- Fully reproducible and aligned with course requirements ✓

Interpretability results should be interpreted with caution

Final Takeaways

- ✓ Transformer-based PLMs encode linguistic information internally
- ✓ Probing is a practical and widely used interpretability method
- ✓ Linguistic knowledge varies across layers and models
- ✓ Small, well-designed prototypes are sufficient for educational analysis

Why probing?→ Because it is the most common and practical method identified in the survey. ✓

Does extraction mean usage?→ No. Extractability does not imply causal reliance. ✓

Why BERT?→ Open architecture, lower computational cost, and widespread use in interpretability studies. ✓

The end

Thank you for your attention